

Outlook markers (as abstract discourse markers): An analysis of *nanka* in Japanese*

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Abstract We propose a formal model for dealing with what we call *outlook markers*, a subtype of discourse markers that govern discourse management pertaining to negotiations about ‘agreeable outlooks’ (on possibly subjective matters). For this purpose, we take up the semantic analysis of an ‘emphatic’ discourse marker *nanka* in Japanese. We argue that *nanka* is essentially a counterstance marker in that it imposes a constraint on the prior discourse that a contextually salient ‘counterstance’ is at issue, and that the speaker’s own stance embodied in the prejacent is incongruent with this counterstance as a subjective judgment.

The model we develop combines Coppock’s (2018) outlook-based semantics for the treatment of subjectivity and Farkas & Bruce’s (2010) Table model for discourse management. We demonstrate that this unified model illuminates the way in which the notoriously elusive evaluative implication of *nanka* arises from the very nature of this particle as an abstract, meta-level discourse marker: it explicitly identifies the discourse conflict as a conflict of in principle irreconcilable opinions (which could threaten the whole point of ‘agreeable outlook’ management).

Keywords: *nanka*, discourse marker, negative evaluation, projective meaning, discourse management, outlook-based semantics, subjectivity

1 Introduction

In this paper, we propose an analysis of the discourse marker *nanka* in Japanese, with a larger goal of developing a formal model for dealing with what we call *outlook markers*, a subtype of discourse markers that govern discourse management pertaining to negotiations about ‘agreeable outlooks’ (on possibly subjective matters). Descriptively, *nanka* is an emphatic marker that invokes a negative/subjective implication whose precise nature is difficult to pin down (Teramura 1991; Sawada 2007; Ido 2023). It typically occurs in contexts such as the following:

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- (1) A: Satoo-iri-no ryokucha-tte oishii-yo-ne.
 sugar-with-GEN green.tea-TOP delicious-SFP-SFP
 ‘Green tea with sugar is delicious, isn’t it?’
 B: E, satoo-iri-no ryokucha-**nanka** oishiku-nai-yo.
 oh, sugar-with-GEN green.tea-NANKA delicious-NEG-SFP
 ‘Oh, green tea with sugar isn’t delicious (at all).’

In (1), B responds to A’s utterance by expressing disagreement. Descriptive studies (see, e.g., Sawada 2007; Ido 2023) take *nanka* to mark B’s ‘negative attitude’ in such a response, but the key function of this particle remains highly elusive.¹

In this paper, we analyze *nanka* as an abstract discourse marker and sketch a formal framework that captures its discourse function adequately. To state the conclusion first, we argue that *nanka* requires the utterance situation to include a conflicting stance (which we call a *counterstance*, borrowing the terminology from Kennedy & Willer 2022) as a candidate for joint context update, and proposes to replace the counterstance with the one expressed by the prejacent. Stances are modelled as outlooks in the sense of Coppock 2018, roughly, propositions expressing (the speaker’s) subjective opinions. The ‘negative’, evaluative implication arises as a derived effect of the relationship between the prejacent and the counterstance, and the fact that these stances reflect underlying irreconcilable opinions.

A novel aspect of the present work lies in the fact that it focuses on the abstract, ‘structural’ aspect of the discourse: in addition to the ‘here’, ‘now’, ‘I’ and ‘you’ of the speech situation,² natural language discourse has a more abstract dimension. It embodies a complex, joint narrative history in which interlocutors interact with one another with the goal of arriving at a consensus. The question then is, does natural language ever mark such abstract aspects of discourse by overt linguistic items?

Our claim is that such items (namely, *outlook markers*) are indeed attested, and that *nanka* is one of them: a class of focus particles in Japanese (including *kurai*, *koso*

¹ All examples of *nanka* in this paper contain morphological negation in the predicate, but this is not a syntactic/semantic requirement. A negatively evaluative predicate such as *mazui* ‘tasteless’ can license *nanka* without overt morphological negation as in (i). Clarifying what this notion of ‘negativity’ amounts to is an important issue, but we won’t attempt to explore it further in this paper.

(i) Satoo-iri-no ryokucha-nanka mazui-ni kimatteiru.
 sugar-with-GEN green.tea-NANKA bad sure
 ‘Green tea with sugar is definitely bad.’

² Previous research on discourse markers has mostly focused on ‘deictic’ aspects of the speech situation such as the speaker’s and hearer’s epistemic states (see, e.g., Davis 2009). But see Deo 2022 for a notable exception, which highlights the function of Hindi *=to* as a general downtoner in discourse.

and contrastive *wa*; Kubota & Ido 2025; Kubota 2025) serve precisely this function. These particles appear to lack substantial meaning, but they serve specific functions as meta-level devices that signal the relationship between the update potential of the current utterance and an at-issue context update candidate (e.g., indicating a direct conflict, proposing to weaken the joint update proposal, etc.).³

Outlook markers reflect fundamental properties of human language, and it is important to develop a formal model that captures their properties adequately. This can be done by combining two independently developed approaches: Farkas & Bruce's (2010) Table model of discourse management, and Coppock's (2018) outlook-based semantics for subjectivity. Our analysis treats *nanka* as a meta-level discourse marker whose function is to signal a conflict between the speaker's (or some attitude holder's) own stance encoded in the prejacent and a contextually salient counterstance that is currently a live candidate for context update.

2 Basic properties of *nanka*

We start by reviewing five key descriptive facts about *nanka*: (i) semantic vacuity, (ii) the necessity of a counterstance in the preceding discourse, (iii) the two-sided nature of the non-at-issue meaning, (iv) embeddability, and (v) the stance-sensitivity of evaluative meaning. Taken together, these properties elucidate the function of *nanka* as an abstract discourse marker, whose primary function is to signal the relationship between the prejacent and the contextually salient counterstance.

2.1 Semantic vacuity and the necessity of a counterstance in the context

One notable property of *nanka* is that it contributes almost nothing substantial to the meaning of the sentence. For example, in (1-B) (reproduced in (2)), replacing *nanaka* in B's response with the topic marker *wa* results in almost no change in meaning:

- (2) B:E, sato-iri-no ryokucha-{*nanka/wa*} oishiku-nai-yo.
oh sugar-with-GEN green.tea-NANKA/TOP delicious-NEG-SFP
'Oh, green tea with sugar isn't delicious (at all).'

Here, B responds to A's claim that green tea with sugar is tasty by taking an opposite stance. The relation between A's and B's utterances is already clear from the context, and *nanka* simply reinforces, as it were, the tension between A's and B's stances. In

³ See Egg (2013) for a closely related view on the discourse function of certain modal particles in German. Our approach can be seen as a further elaboration of the ideas prefigured in Egg's work which works out the interactions of subjectivity and discourse update management more explicitly in a formal model of discourse.

this sense, *nanka* can be seen as a purely pragmatic device—its sole function is to explicitly mark the discourse relation between utterances.

Previous studies have noted that *nanka* typically requires the presence of a proposition in the prior discourse that contrasts with, or is inconsistent with the prejacent (Sawada 2007; Ido 2023), namely, a counterstance (cf. (2) above).

When no such counterstance is present, *nanka* is unacceptable. For example, by slightly modifying the exchange in (1) by replacing A’s original utterance by a general *wh*-question, B’s response becomes infelicitous. In (3), whether or not sweetened green tea is delicious is not at issue immediately prior to B’s utterance. Such a context does not support the use of *nanka*, as there is no salient counterstance.

- (3) A: Donna nomimono-ni satoo-o ireru-no-ga oishii-ka-na?
 what.kind drink-DAT sugar-ACC put-GEN-NOM delicious-Q-SFP
 ‘I wonder what kind of drink tastes good with sugar in it.’
 B: Satoo-iri-no ryokucha-{wa/??nanka} oishiku-nai-yo.
 sugar-with-GEN green.tea-TOP/NANKA delicious-NEG-SFP
 ‘Sweetened green tea is not tasty (at all).’

2.2 The two-sided nature of the non-at-issue meaning

The requirement of the existence of a counterstance is a presupposition. *Nanka* additionally invokes a negative evaluation meaning, which is also arguably not part of the truth-conditional content of the sentence—most likely some type of Pottsian CI (Potts 2005) similar to expressives (but see section 2.3 below on embeddability). Of the two types of non-at-issue meaning, the presuppositional status of the counterstance requirement is perhaps less problematic—this can be tested by using the adverbial expression *somo-somo* ‘to begin with’, which is an expression that questions the premise (Kubota 2016). As in (4), the counterstance can be denied with the use of *somo-somo*, which indicates that it is a presupposed, rather than asserted, content:

- (4) A: Satoo-iri-no ryokucha-**nanka** Nihonjin-no Tanaka-san-wa
 sugar-put-GEN green.tea-NANKA Japanese-GEN Tanaka-Mr.-TOP
 noma-nai yo.
 drink-NEG SFP
 ‘Mr. Tanaka, being Japanese, won’t drink green tea with sugar or anything like that.’
 B: Uun to, somosomo ore-wa kare-ga soo iu mono-o nomu
 Uh SFP to.begin.with I-TOP he-NOM that.kind thing-ACC drink

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kamo to-wa itte-nai zo.
may COMP-TOP say-NEG SFP

‘Uh, to begin with, I never said he might drink that kind of thing.’

The negative evaluative meaning is much harder to pin down precisely, but one property that it shares with other non-at-issue meanings is that it cannot be the target of a yes/no response by the hearer. Thus, as shown in (5), when B responds to a sentence uttered by A containing *nanka* with a simple ‘no’ response, what is denied is the propositional content of A’s utterance rather than whatever additional negative meaning is invoked by the use of the word *nanka*.

- (5) A: Satoo-iri-no ryokucha-**nanka** watashi-wa noma-nai.
 sugar-put-GEN green.tea-NANKA I-TOP drink-NEG
 ‘I won’t drink green tea with sugar or anything like that.’
B: Iya, sonna hazu-wa-nai yo.
 No such case-TOP-NEG SFP
 ‘No, that can’t be true.’
 (≈ ‘You’ll drink it for sure.’)
 (≠ ‘You are not negative about green tea with sugar.’)

Another observation we can make is that this meaning component has the property of ‘descriptive ineffability’ and ‘immediacy’, which are identified as typical properties of expressives by Potts (2007). That is, the negative connotations associated with outlook markers are manifestations of attitudes or stances that directly result from the use of particular words (that is, they are ‘immediate’), and hence are not something that can be paraphrased by propositions that are either true or false (that is, they are ‘ineffable’).⁴

⁴ We can perhaps make this intuition somewhat sharper by considering a context in which one loses precisely the ability to express such attitudes or stances. Suppose John, an eminent athlete who is desperately trying to overcome the pressure of preparing for the next Olympics, has decided to participate in a clinical trial of a new drug which has the effect of giving one the ability to monitor one’s own emotional states completely and to report them accurately in language. But this drug has a side effect: it deprives the subject of the ability to *express* emotions and attitudes through the use of language. The drug worked extremely well for John, but the side effect was also severe. Having taken the drug as usual and feeling that its effect is now in full force, John says:

- (i) a. Watashi-ni-wa kin-medaru-wa tor-e-nai. Watashi-wa jishin-o ushinat-te
 I-DAT-TOP gold-medal-TOP get-POT-NEG I-TOP confidence-ACC lose-TE
 i-ru.
 PROG-NPST
 ‘I am not going to win a gold medal. I have lost confidence.’

2.3 Embeddability

As noted above, the negative evaluative meaning of *nanka* is similar to the expressive meanings conveyed by honorifics and slurs, in that it reflects the attitude of some agent. However, there is one crucial difference between *nanka* and typical expressives. While expressive meanings are generally strictly speaker-oriented, the negative meaning of *nanka* can shift its orientation to a local attitude holder in certain embedded environments, most typically under attitude predicates (Ido 2023).

For example, (6) can be uttered in a context in which the speaker was confident that their paper would be accepted but their advisor held a more pessimistic view.

- (6) Sensei-wa [boku-ga SALT-ni-**nanka** toora-nai]-to omot-te-ta rashii.
 advisor-TOP I-NOM SALT-DAT-NANKA pass-NEG-C think-PST seem
 ‘My advisor seems to have thought I wouldn’t possibly get accepted at SALT.’
 ⇨ The advisor had a negative/pessimistic outlook.

Here, *nanka*’s evaluative meaning can only be read as reflecting the advisor’s stance.

2.4 The stance-sensitivity of the negative evaluation meaning

The ‘negative evaluation’ meaning of *nanka* is expressive-like but is more abstract since it is embeddable under attitudes. It arguably arises as a by-product of the relationship between the counterstance and the prejacent, but its exact nature is rather elusive. In this final subsection, we hone in on this issue somewhat more closely: we will see below that the negative evaluation meaning of *nanka* is fundamentally subjective: it arises from the very fact that the pragmatic tension between the prejacent and the counterstance signalled by *nanka* is in turn based on the fact that the truth of the prejacent depends on the stance one takes to support the claim.

The prejacent of *nanka* expresses a subjective judgment. The following examples illustrates this point clearly.

- (7) A: These shishitoo peppers look spicy.
 B1: [B has eaten shishitoo pepper many times before, and in their own experience and given their standard for spiciness for food, shishitoo peppers don’t count as ‘spicy’.]

W: Watashi-ni-wa kin-medaru-**nanka** tor-e-nai.
 I-DAT-TOP gold-medal-NANKA get-POT-NEG
 (un)intended: ‘I *can*’t possibly win a gold medal.’

(ia) can be understood as an utterance by John under the influence of the drug, but (ib) is simply uninterpretable. The context makes it clear that John is in a state in which he is *incapable* of using an expression like *nanka* felicitously.

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Iya iya, shishitoo-{nanka/wa} karaku-nai-yo.
no no shishito-NANKA/TOP spicy-NEG-SFP

‘No, no, shishitoo peppers aren’t spicy.’

B2:[After chacking some reliable sources online:]

Uun, shishitoo-{??nanka/wa} karaku-nai-yo. Capsaicin sukunai-kara.
uh shishito-NANKA/TOP spicy-NEG-SFP capsaicin low-because

‘Uh, shishitoo peppers aren’t spicy—they contain little capsaicin.’

In (7-B1), B1 responds to A’s question by reporting their own perceptual experience. In this type of context, *nanka* is perfectly natural. By contrast, in (7-B2), B2 doesn’t have any direct experience of shishitoo pepper’s spiciness, but bases their judgment solely on factual information about its capsaicin content. In this latter case, the context makes it clear that *karai* ‘spicy’ doesn’t refer to personal taste, but agricultural facts about the particular kind of pepper that has low content of capsaicin. And this fails to license *nanka*.⁵

Another type of judgment that is in many cases inherently subjective (but for a somewhat different reason) is modal judgment. In modal sentences, truth judgment depends on the assumptions and background considerations that serve as the basis for making the judgment (this is given by the modal base and the way it is structured by the ordering source in Kratzer’s (1981) framework). As shown below, *nanka* can occur naturally in modal sentences, with some interesting and subtle variation in the way in which its negative evaluation implication is insinuated.

- (8) a. Konna tokoro ni nihonjin-nanka inai hazu da.
such place LOC Japanese-NANKA exist-NEG supposed COP
‘There shouldn’t be any Japanese people in a place like this.’
- b. Watashi ni-wa kin medaru-nanka torenai.
I DAT-TOP gold medal-NANKA get-NEG
‘There’s no way I can win a gold medal.’
- c. Poruno shoosetsu-nanka yomu beki de-wa nai.
porn novel-NANKA read should COP-TOP NEG
‘You shouldn’t read things like porn novels.’

⁵ (7-B2) improves in a situation in which B2 attempts to claim epistemic advantage over A, often implying a contempt on A’s lack of knowledge. Such cases can be thought of as involving an additional layer of epistemic modality. Once we admit this meaning shift, *nanka*’s felicity can be explained in the same way as with epistemic modals (see section 3.4.1). Note that even (ostensibly) episodic past sentences such as *Taroo-nanka konakatta-yo* ‘Taro didn’t come (what are you talking about?)’ can be targeted by *nanka*. But such examples are no longer neutral reports of past events but are felicitous only on the extended, epistemic reading.

- d. Aisu kurmu-nanka iranai.
ice cream-NANKA need-NEG
'I don't need something like ice cream.'

For example, in (8a), which involves epistemic likelihood or stereotypicality, truth judgment depends on such factors as the nature of the area in question (e.g., its safety), the typical behavior of the Japanese, or factual information such as eyewitness, and the way in which these factors are weighted. In (8c) involving deontic modality, by contrast, the relevant factors can be differences in moral standards, understanding of social norms and allowable deviations from such norms, etc. All of this can naturally vary from individual to individual.

In modal sentences with *nanka*, the conflict between the prejacent and the contextually at issue counterstance is supported by the truth variability of the judgment, reflecting the different background assumptions which in turn are rooted in deeper differences in stance among different individuals. When epistemic likelihood/stereotypicality or ability estimate is what is at issue (as in (8a) and (8b)), truth variability just reflects different understandings of such (more or less) neutral matters. By contrast, when what is at issue is moral judgment or personal preference/desire (as in (8c) and (8d)), truth variability reflects underlying core values and preferences that different individuals bring in when making value-laden judgments. Signalling a tension between one's own stance and a contextually salient counterstance explicitly in such cases almost inevitably invokes adversarial or pejorative implications.

Interestingly, in modal sentences too, the acceptability of *nanka* varies depending on whether truth judgment depends on an objective standard or not.

- (9) A:Kono anken, hojokin-no taishoo-ni naru?
this case subsidy-GEN eligible-to become
'Does this project qualify for the subsidy?'
B1:Eeto, kitei-ni-yoruto sakunendo-chakkoo-no baai-wa
uh regulation-according-to last.year-start-GEN case-TOP
hojokin-shinsei-{??nanka/wa} dekimasen.
subsidy-application-NANKA/TOP can.not
'According to the regulations, a subsidy isn't applicable for a carry-over project.'
B2:Sakunendo-chakkoo-no baai-wa hojokin-shinsei-{nanka/wa}
Last.year-start-GEN case-TOP subsidy-application-NANKA/TOP
dekimasen. Tadade-sae yosan-ga tarinai-node.
can.not as.it.is even budget-NOM insufficient-because
'There's no room for subsidy application for a carry-over project. The budget is already tight.'

Both sentences in (9) express judgments about whether subsidy is available, but different types of criteria are involved in supporting the judgment. (9B1) reflects an objective judgment, based on the relevant regulations, which is invariant regardless of whatever stance one takes. By contrast, (9b) embodies a subjective judgment, where the relevant criterion is one’s assessment about the current budget situation. Underlying considerations supporting such judgments can be highly subjective—some people make optimistic assessments, while others are more pessimistic or realistic; the way in which different people predict the future can also make a difference in the way they read the budget situation, etc. This gives rise to stance-dependent truth variability. As can be seen in the contrast in (9a,b), the felicity of *nanka* is dependent on the presence/absence of this truth variability.

3 Analyzing *nanka* as an abstract discourse marker

We have seen that *nanka* imposes two constraints: (i) the prejacent must be a subjective proposition; (ii) a counterproposal must be at issue as a current candidate for context update. To capture these two properties, we need a framework that models natural language discourse as a system of negotiation among interlocutors over subjective propositions. For this purpose, we combine Coppock’s (2018) outlook-based semantics as a theory of subjective propositions and Farkas & Bruce’s (2010) Table model as a theory of discourse update. The resultant model allows us to pin down the source of the elusive ‘negative evaluation’ meaning tied to *nanka* precisely (or, at least, more precisely than our precursors) without explicitly encoding such ‘subjective’ implication as a lexically encoded non-at-issue meaning of *nanka*.

To deal with the embeddability properly, a minor, but important theoretical extension is needed: we extend the Table model so that it can deal with embedded Tables, which track ‘previous discourse’ of embedded thought. While the notion of an embedded Table may seem exotic, a hallmark of abstract discourse markers is that they are sensitive to aspects of context update not directly tied to the utterance situation itself. Embedded properties of *nanka* (and similar ‘subjective’ particles and adverbs in Japanese; cf. Kubota 2025) thus call for assuming such a notion, as a natural extension of the already available notion of embedded context for presupposition accommodation in dynamic semantics (cf., e.g., Heim 1992).

3.1 Formilizing the counterstance requirement in the Table model

The Table model (Farkas & Bruce 2010) is characteristic for explicitly modelling pending issues in discourse update. That is, the Table is a stack of propositions currently under negotiation among interlocutors as candidates for CG (or Context Set) update. This architecture is suitable for analyzing *nanka*, as it is essentially a

device that signals the relationship between the prejacent and the status of the Table.

Within the Table model, we can explicitly represent the relationship between the prejacent of *nanka* and the contextually salient counterstance. The first (simplified) version of the meaning of *nanka- ϕ* , where ϕ is the prejacent, is shown in (10).

$$(10) \quad \textit{nanka-}\phi = \lambda \langle C, T \rangle. \partial (\exists \phi' \in T. \phi' \cap \phi = \emptyset); \langle C, T[\phi' / \phi] \rangle$$

Here we assume that an utterance is a function that takes a tuple of a Context Set (CS) C and a Table T as its input and returns an updated CS-Table tuple as its output. In (10), *nanka- ϕ* leaves the input CS C unchanged. That is, *nanka*-sentences have a vacuous context update effect: it merely affects the Table, and in a rather trivial manner. Specifically, *nanka* replaces a proposition ϕ' that is already on the Table with the prejacent ϕ . It additionally requires (in the definedness condition, i.e., as a lexical presupposition) that the counterstance ϕ' and the prejacent ϕ be incompatible, that is, they cannot both be true simultaneously. In other words, *nanka* is essentially a counterstance marker: it overwrites a current context update candidate with the speaker's own remark, signalling explicitly the contradiction between the two. Effectively, *nanka* requires its prejacent to directly respond to the already salient current issue, without introducing any new information. This way, the analysis in (10) captures the semantic vacuity noted in section 2 straightforwardly.

The existence requirement of a counterstance on the Table directly accounts for the infelicity of *nanka*-sentences in contexts such as (3), repeated as (11), in which *nanka* is used in a response to a general *wh*-question. Here, the condition in (10) is not satisfied since there is no salient counterstance in the prior discourse.

- (11) A: I wonder what kind of drink tastes good with sugar in it.
 B: Satoo-iri-no ryokucha-{wa/??nanka} oishiku-nai-yo.
 sugar-with-GEN green.tea-TOP/NANKA delicious-NEG-SFP
 ‘Sweetened green tea is not tasty (at all).’

(10) is still inadequate in that it does not capture the fact that *nanka*-sentences express subjective propositions, nor does it (directly or indirectly) encode the speaker's ‘negative attitude’. Note that mere logical contradiction encoded in (10) cannot by itself account for the emergence of such a negative evaluative implication—after all, it is perfectly possible to objectively and logically counter an opposing view without expressing any affective stance. But such purely logical argument is by far the least appropriate environment for *nanka* (see also section 3.4.2 for a related discussion).

Thus, (10) only represents one aspect of *nanka*'s meaning. The evaluative implication is thought to arise from two interacting factors: (i) the requirement that the prejacent ϕ be a subjective proposition, whose truth may vary depending on the stance one takes; and (ii) the discourse-level opposition between counterstances,

reflecting the conflict between the speaker’s own stance and the one currently at issue. In section 3.2, we refine this analysis by formalizing the relationship between the prejacant and the counterstance as an opposition between subjective propositions. This results in a more complete account of the ‘subjective’ meaning of *nanka*.

3.2 Subjectivity in outlook-based semantics

Outlook-based semantics can be seen as a fine-grained version of truth-conditional semantics in which subjective propositions have a natural ontological place: they are just propositions, in the suitably refined sense. More specifically, in outlook-based semantics, outlooks are refinements of possible worlds—different outlooks reflect underlying fine-grained judgments about evaluative/normative standards or preferences about sensory experiences, that is, matters that don’t pertain to objective truth. Once the world is settled, the truth of all objective propositions are fixed; once the outlook is settled, the truth of all subjective propositions are fixed in addition. In this setup, more than one outlook can correspond to a single world, and one can define a set of equivalence classes, called ‘refinement classes’ of outlooks that stand in one-to-one relations to the members of the set of worlds. The members of the same refinement class are entailment-preserving with respect to propositions that describe objective facts (see the definition in (12a) below). Propositions that reflect subjective opinions intuitively are called *discretionary propositions*. These are sets of outlooks that ‘make a cut’ across worlds (i.e., their truth can vary depending on outlooks in at least one world). Standard notions such as entailment and contradiction are defined in the familiar manner, modulo the replacement of possible worlds by outlooks.

There are several alternative approaches to subjectivity. A major alternative is one that introduces the judge parameter explicitly in the semantic representation (see, e.g., Stephenson 2007b). The two approaches are for the most part notational variants, but there are a couple of conceptual advantages that outlook-based semantics offers for the purpose of analyzing abstract discourse markers such as *nanka*. Most importantly, in outlook-based semantics all propositions are indiscriminately treated as sets of outlooks, and the distinction between objective and subjective (or discretionary) propositions is determined solely by the properties of these sets (see (12)). There is no type distinction, nor is there any additional (bound or free) variables in the semantic object that models propositions. In other words, in this approach, natural language propositions are inherently objects that (in principle) reflect subjective judgements. An ‘objective proposition’ is simply a special case—a set of outlooks that satisfy the particular constraint in (12a) that makes it invariant across ‘judges’.

Technically, objective propositions and discretionary propositions are distinguished by the properties they respectively satisfy, along lines spelled out in (12).

- (12) a. A set of outlooks O is an *objective proposition* if and only if there is some subset W of $\mathcal{W}$ such that $O = \bigcup \{O' \subseteq \Omega \mid O' \propto w \text{ for some } w \in W\}$.
 b. A set of outlooks O is a *discretionary proposition* if and only if it is not an objective proposition.

$\mathcal{W}$ and Ω are sets of all worlds and all outlooks in the model, respectively. $\propto$ is the correspondence relation between a set of outlooks and a world that holds if and only if the former is a refinement class for the latter. What (12a) says is that for any given world w , O either contains all of its refinement outlooks or doesn't contain any of them. In other words, there are no world-mate outlooks o_1 and o_2 such that $o_1 \in O$ and $o_2 \notin O$. That is, for any given world w (and its refinement class $O' \propto w$), the truth of O is invariant since it is either true or false uniformly for all members of O' .

This architecture is optimal for our purposes for two reasons. First, once extended dynamically, it offers a succinct view of discourse as a system of negotiation over agreeable outlooks, where an outlook embodies both understandings of objective facts and judgments pertaining to subjective matters. Second, in this approach, attitude embedding corresponds to anchoring a subjective judgment to the attitude holder's outlook. These two features jointly make it possible to capture the properties of *nanka* as an outlook marker in a conceptually coherent and formally natural way.

We revise the analysis of *nanka* as in (13) within outlook-based semantics, where **discretionary**(φ) means that φ is a discretionary proposition in the sense of (12b).

$$(13) \quad \text{nanka-}\varphi = \lambda \langle C, T \rangle. \partial (\text{discretionary}(\varphi) \wedge \exists \varphi' \in T. \varphi' \cap \varphi = \emptyset); \langle C, T[\varphi' / \varphi] \rangle$$

The only difference from (10) is that (13) explicitly requires the prejacent φ to be discretionary. This seemingly trivial revision has two important consequences. Let us check the simpler one first. (13) predicts the contrasts in (7) and (9), where *nanka*'s acceptability is affected by whether the prejacent counts as an objective proposition in the sense of (12a). For example, in (7-B2), the prejacent refers to encyclopedic knowledge. Because this yields a proposition whose truth solely depends on objective facts, it violates the condition in (13). The contrasts in the modal examples in (9) follow in a similar manner.

The other consequence pertains to the source of the negative evaluation implication, and the logic behind this is somewhat more involved. Essentially, our claim is that this implication arises from the fact that *nanka* signals a conflict between counterstances *qua* discretionary propositions. We have noted above in connection to (10) that the mere fact that the prejacent contradicts a contextually salient update proposal doesn't by itself invoke any negative meaning. What is crucial here is that (13) encodes something more specific: not only is there a conflict, but the conflict pertains to a subjective matter. The choice to update the context with *nanka*-marking then implies that this strategy is in some sense more 'appropriate' than the unmarked,

more general update strategy (without *nanka*) in which the contradictory proposition φ' on the Table is simply thrown away quietly by a default process of eliminating inconsistency. The virtually only reason to prefer this marked strategy over the default one is to make it explicit that the conflict between the prejacent φ and the counterstance φ' arises from the very fact that they are discretionary. A discretionary proposition is one whose truth may vary even after settling all factual understandings. In other words, when two propositions contradict with each other because they are subjective, that means that the disagreement in judgment stems not from different understanding of objective facts but from a purely subjective difference in stance.

Put simply, *nanka* encodes that the conflict is not at the level of worlds but at the level of outlooks, reflecting fundamental subjective stances. But if this conflict is not just subjective but is based on personal preferences or core value judgments (on desires, social norms, etc.), reaching a common outlook by rational means is in principle impossible. The act of insisting on overwriting a pre-existing consensus-making candidate with one's own stance that contradicts with it then amounts to rejecting the counterstance as something that is out of question beyond reason. Due to the nature of the discourse move this entails, this gives rise to the strong derogatory or pejorative implications associated with *nanka* in such contexts.

3.3 Embedding outlooks

We have seen that *nanka* is a counterstance marker. It requires the presence of a counterstance, and, moreover, the incompatibility between the pre-existing counterstance and the new update proposal is not merely about objective facts but reflects conflicting outlooks on subjective matters. In principle, stance marking is relevant in any situation involving belief updates; this includes not just actual conversational interactions, but also reported beliefs expressed by propositional attitude predicates.

Under outlook-based semantics, embedded propositions are themselves objects that reflect (in principle) subjective judgments, that is, they are sets of outlooks. Then, just as discourse update is a dynamic process of negotiation of agreeable outlooks, belief update can be thought of as a process in which the attitude holder dynamically chooses to take a stance in relation to contextually relevant 'counterstances' (i.e., competing candidates for belief update that are present in the attitude holder's mind).

To model this situation explicitly, we need the notion of an embedded Table, and an explicit analysis of stance-taking dynamics in belief update. This can be achieved by refining the dynamic analyses of attitude embedding such as Heim 1992. Essentially, embedded *nanka* targets the local context associated with the belief state of the attitude holder, in a way parallel to the context update effect of locally accommodated presuppositions—just as interlocutors take stances in context update negotiations in actual conversation contexts, the belief holder manages the

‘negotiations’ between competing outlooks in their mental discourse.

For this purpose, we extend doxastic alternatives to come with their associated Tables. Specifically, we define ‘Table-extended doxastic alternatives’ $\mathbf{DOX\&T}_{x,t,o}$ for agent x at time t in outlook o , which is a tuple of doxastic alternatives (in the standard sense) and a Table, the latter of which is a set of propositions that the agent hasn’t yet accepted. Then, we can define the meaning of *kangaeru* ‘believe’ as in (14), where $\mathbf{believe}(p, x, t)$ (modelled after Heim 1992 with adjustments) is a set of outlooks in which the doxastic alternatives for x at t (dynamically) changes from the immediately preceding time t_0 such that they entail p (i.e., the CS component of $\mathbf{DOX\&T}_{x,t,o}$ entails p).⁶

$$(14) \quad \mathbf{believe} = \lambda p \lambda x \lambda t \lambda o. \partial(\exists t_0 \prec t \wedge \mathbf{DOX\&T}_{x,t_0,o} \in \mathbf{Dom}(p)).$$

$$\mathbf{DOX\&T}_{x,t,o} = \uparrow(\mathbf{DOX\&T}_{x,t_0,o} + p)(p)$$

where $\uparrow(C)(p) = \langle \mathbf{CS}(C) \cap p, \mathbf{TBL}(C)[p/\top] \rangle$

In (14), $\uparrow$ denotes a CS update function such that $\uparrow(C)(p)$ moves p from the Table of C to its CS (replacing p with the tautology $\top$ (a vacuous claim) is the same as removing it from the Table).⁷

The definedness condition in (14) requires that the doxastic alternatives $\mathbf{DOX\&T}_{x,t_0,o}$ at the pre-state of belief formation is in the domain of the update function p (i.e., for an attitude holder to sensibly ascribe to themselves the believe p , their doxastic alternatives at the pre-state has to allow for the possibility of updating with p). This plays a crucial role in the way the felicity condition on embedded *nanka* plays out in the interpretation of the sentence.

Let’s unpack this with the example in (15), whose translation is given in (16).

- (15) Sensei-wa [boku-ga SALT-ni-{nanka/wa} tooranai] to omotta.
 advisor-TOP I-NOM SALT-DAT-NANKA pass.NEG C think.PST
 ‘My advisor thought that I wouldn’t possibly get accepted at SALT.’
 $\rightsquigarrow$ The advisor had a negative/pessimistic outlook on the outcome of the speaker’s SALT submission.

$$(16) \quad (15) = \mathbf{believe}(\varphi)(a)(t)$$

$$= \lambda o. \partial(\exists t_0 \prec t \wedge \mathbf{DOX\&T}_{a,t_0,o} \in \mathbf{Dom}(\varphi)).$$

$$\mathbf{DOX\&T}_{a,t,o} = \uparrow(\mathbf{DOX\&T}_{a,t_0,o} + \varphi)(\varphi)$$

where $\varphi = \mathbf{NANKA}(\mathbf{get-accepted-at-salt}(\mathbf{speaker}))$
 $a = \text{‘the speaker’s advisor’}$

⁶ For convenience, we take immediate precedence $\prec$ to be an unanalyzed primitive.

⁷ $\mathbf{CS} = \pi_1$ and $\mathbf{TBL} = \pi_2$; mathematically these are just projection functions which return the first (CS) and second (Table) components of a CS-Table tuple, respectively.

The definedness condition imposed on **believe** in (16) contains the clause saying that updating $\mathbf{DOX\&T}_{a,t_0,o}$ (the doxastic alternatives for the advisor at t_0 in o) with φ is well-defined, that is, the former is in the domain of the latter. Thus, (16) can be further unpacked as in (17).

(17) a. **definedness condition:**

$$\mathbf{discretionary}(\varphi) \wedge \exists \varphi' \in \mathbf{TBL}(\mathbf{DOX\&T}_{a,t_0,o}). \varphi' \cap \varphi = \emptyset$$

b. **denotation:**

$$\begin{aligned} \lambda o. [\mathbf{DOX\&T}_{a,t,o} &= \uparrow(\mathbf{DOX\&T}_{a,t_0,o} + \varphi)(\varphi)] \\ &= \lambda o. [\mathbf{DOX\&T}_{a,t,o} = \langle \mathbf{CS}(\mathbf{DOX\&T}_{a,t_0,o}) \cap \phi, \mathbf{TBL}(\mathbf{DOX\&T}_{a,t_0,o})[\varphi' / \top] \rangle] \end{aligned}$$

(17a) says that at a time immediately before the belief update takes place, a conflicting claim φ' was on the Table for the advisor's thought as a potential candidate for belief update. (17b) replaces this counterstance on the Table with the prejacent φ , and updates the CS with the latter.

Crucially, the recognition of inconsistency between φ and φ' is within the local attitude holder's thought and not attributed to the speaker. In other words, embeddability of *nanka*'s meaning follows from the fact that this meaning component pertains to the modulation of CS update in conflict 'narrative situations' for the belief holder in the case of embedded thought.

At this point, one may wonder what it means to say that a 'counterstance' exists in belief update. To see what this means, note first that the context relevant to belief updates can be formed by evidence obtained in various ways, including at least direct experience and report from somebody else (e.g., *Jane heard from Mary that Peter would come to the party. She then thought that his wife would come too*). It is then not unreasonable to assume that the context includes not only such background knowledge but also possible stances to take on the relevant matters. The embedded Table can be seen as a theoretical device that represents precisely such alternative stances/outlooks.

This is reflected in the empirical distribution of embedded *nanka* in just the way our analysis predicts: *nanka* in embedded environments is felicitous only when the belief holder is aware of the counterstance that contradicts the prejacent. Thus, the felicity of (15) is affected by the different contexts given in (18), in a way parallel to how the prior discourse affects the felicity of matrix *nanka* (cf. (3)).

(18) **Context 1** [(15) OK with either *nanka* or *wa*]: The advisor was aware of the speaker's confidence in getting accepted at SALT, and was highly skeptical.

Context 2 [(15) OK only with *wa*]: The advisor wondered generally which conference may be good for the speaker.

In Context 1, the advisor's pessimistic stance is explicitly contrasted with a contextually salient counterstance (which happens to be the speaker's own). This satisfies

the licensing condition on *nanka*. By contrast, in Context 2, the advisor is simply thinking generally and widely about possible submission venues. In this case, there is no explicit counterstance to the prejacent stance, and the use of *nanka* is infelicitous. In other words, the acceptability of embedded *nanka* depends on whether the belief context contains a salient counterstance—a competing outlook or judgment that is a ‘live candidate’ for belief update when the local update by the prejacent is enacted.

3.4 Some further thoughts on subjectivity

3.4.1 Epistemic modals

In Coppock’s (2018) definition (see (12a)), objective propositions are those whose truth is determined by purely objective facts, and discretionary propositions as those whose truth is not fixed solely by such facts. Predicates of personal taste and moral judgments involving deontic modals are clearly discretionary since their truth judgments can diverge from one outlook to another even after all relevant facts have been agreed upon (and these predicates pass the ‘faultless disagreement’ test). Epistemic modals have a somewhat delicate place in this classification. Coppock takes them to be objective propositions on the grounds that epistemic judgments are about facts about the world rather than subjective attitudes toward them.

But it does not seem entirely uncontroversial that epistemic modals are just about facts. For one thing, there is no consensus in the literature on whether they pass the ‘faultless disagreement’ test (see, e.g., Stephenson 2007a for a view that they align with taste predicates in this respect). Moreover, as already noted in section 2, *nanka* can naturally occur in examples with epistemic modals such as (19).

- (19) John-wa biiru-**nanka** nom-anai-hazu-da.
 John-TOP beer-NANKA drink-NEG-supposed-COP
 ‘John wouldn’t possibly drink beer.’

We assume (contra Coppock 2018) that epistemic modals can express discretionary propositions and that it is precisely such uses of epistemic modals that license *nanka*. To see this, note that (19) is typically uttered in a context in which the speaker makes their stance explicit when there are conflicting views about whether John drinks beer. In such a situation, the judgments that take (19) to be true or false are each grounded in different types of evidence. Such evidence may include specific knowledge about John (such as his past behavior, age, etc.) and more general, socio-cultural/stereotypical knowledge (whether beer is popular for people in John’s generation, to what extent drinking is taken to be a common practice for adults, etc.).

These matters (e.g., John’s age) are mostly simple facts about the world, but there is ample room for subjective opinion as to *how* such facts are brought to bear on the

truth judgment of the modal statement. Thus, even in epistemic modal sentences, the truth of the overall judgment can vary depending on how the evidential basis is formed. In this sense, we take epistemic modal judgments to be no different from other types of modal judgments in embodying discretionary propositions.⁸

3.4.2 Apparent counterexamples to the discretionarity requirement

Epistemic modals aside, there are cases that seem to pose a more blatant problem for the discretionarity requirement encoded in (13). This is illustrated by (20), in which B apparently reports a simple fact pertaining to an agricultural sortal distinction:

- (20) A:Saikin-no tomato-tte amakute kudamono mitai-da-yo-ne.
recent-GEN tomato-TOP sweet fruit like-COP-SFP-SFP
‘Tomatoes are so sweet these days—they’re like fruit, aren’t they?’
B:Nani itte iru-nda-yo. Tomato-**nanka** kudamono-ja-nai-yo.
what say PROG-COP-SFP tomato-NANKA fruit-COP-NEG-SFP
‘What are you talking about? Tomatoes aren’t fruit (at all).’

Given the commonsense understanding about tomatoes, the truth of (20B) seems objectively determined, leaving no room for subjectivity (or outlook sensitivity).

However, even sentences such as (20B) which appear to report plain factual information can exhibit truth variability depending on the criteria of judgment. In fact, while tomatoes are botanically classified as fruits, there was a 19th-century lawsuit in U.S. Supreme Court that debated precisely the issue raised in (20B): whether tomatoes should be considered fruit or vegetable. The court ultimately ruled that tomatoes should be considered vegetables, reasoning that they are served as part of the main course and not as dessert, etc., thereby overriding the botanical definition in favor of common usage. This episode⁹ illustrates that the truth of a seemingly objective proposition such as ‘tomatoes are vegetables’ depends not solely on scientific facts, but is potentially affected by social and cultural factors.

Nanka is again sensitive to whether the criterion is objective. In (21), the context makes it clear that B bases their judgment on purely scientific botanical facts rather than socio-cultural conventions. In this case, the use of *nanka* becomes infelicitous.

⁸ In Coppock 2018, the objective/discretionary distinction is taken to correlate with the distribution of the belief predicate *tycka* in Swedish, whose complement clause allows only ‘subjective’ propositions, and epistemic modals are banned from this environment. This suggests that the notion of subjectivity relevant to *tycka*’s complement is more restrictive than the type of subjectivity imposed on *nanka*’s prejacents. This raises an interesting issue, but exploring it further is beyond the scope of this paper.
⁹ Perhaps best known to semanticists via Dowty 1979: 387–388, who mentions this case in connection to a discussion on word meaning and foundations of truth-conditional semantics.

- (21) A:Speaking of vegetables, tomato comes to mind.
 B:[Checking relevant facts in an encyclopedia:]
 Uunto, demo shokubutsugaku-teki-ni-wa, tomato-{ ??nanka/wa }
 hmm but botanically-ADV-TOP tomato-NANKA/TOP
 yasai-ja-nai-yo.
 vegetable-COP-NEG-SFP
 ‘Huh. Well, botanically speaking, tomato isn’t a vegetable.’

4 Conclusion (and outlook)

We have proposed an analysis of *nanka* as an outlook marker in a model of discourse combining Coppock’s (2018) outlook-based semantics with Farkas & Bruce’s (2010) Table model. An outlook marker is an abstract discourse marker that signals the relationship between the prejacent as a subjective proposition and a contextually salient counterproposal for discourse update. We believe that outlook markers are pervasive in language. Potential candidates (not meant to be exhaustive) include:

- (22) a. ‘evaluative’ particles and adverbs in Japanese: *kurai*, *koso*, *made-(mo)*, and contrastive *wa* (Ido 2023; Kubota & Ido 2025; Kubota 2025)
 b. particles/adverbs in other languages: *doch* (German; Egg 2013; Repp 2013), *überhaupt* (German; Rojas-Esponda 2014), *=to* (Hindi; Deo 2022)
 c. scale-sensitive expressions: *even* (Greenberg 2018), *at least* (Biezma 2013; Chen 2024), *almost/barely* (Horn 2017)

The proposed model takes outlook update management to be the key property of natural language discourse, working out concretely an idea prefigured in Coppock 2018: §4. Outlook markers have a natural place in this model since they are meta-level devices that signal explicitly what type of outlook update is being proposed. Outlooks and outlook update management can moreover be embedded, since natural language can report embedded thought. This model may offer a new, unified outlook on the complex behaviors of phenomena extensively studied in the semantics literature such as those in (22).

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